

ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination questions
January 2018

Instructions: Closed book. NO CALCULATORS are allowed. This examination contains **seven** problems, each worth 20 points. Do any **five** of the seven problems. Show your work. Clearly indicate which **five** are to be graded.

PLEASE GRADE PROBLEMS: 1 2 3 4 5 6 7

1. Consider the Newton-Raphson method with given initial guess x_0 used to generate a sequence $\{x_n\}$, $n > 0$ to calculate the reciprocal of the number N by solving $\frac{1}{x} = N$.
 - (a) Show that the iteration can be written as $n \geq 0$, $x_{n+1} = x_n(2 - Nx_n)$.
 - (b) Show that if the sequence $\{x_n\}$ converges to the root, then the convergence rate is *quadratic*.

2. (a) Given n distinct data values (t_i, y_i) , $i = 1, \dots, n$, $t_i \in \mathbb{R}^1$, $y_i \in \mathbb{R}^1$, $t_1 < t_2 < \dots < t_n$, show that there exists a unique polynomial of degree at most $n - 1$ that interpolates this data. Describe a method to find this polynomial.
 - (b) Let $P_1(f; t)$ be a polynomial of degree 1 interpolating the function $f(t)$ at t_1, t_2 . If $f \in C^2[t_1, t_2]$, show that the error in linear interpolation satisfies,

$$f(x) - P_1(f; t) = (t - t_1)(t - t_2) \frac{f''(\xi(t))}{2}, \quad t_1 < t < t_2$$

3. Consider solving the linear system of equations $Ax = b$ for an invertible matrix A and a fixed right hand side $b \neq 0$.
 - (a) Given an approximate solution $\hat{x} \neq 0$, set $\hat{b} = A\hat{x}$ and define the relative forward error (RFE) and relative backward error (RBE) for this problem.
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 - (c) Compare the condition number of the above problem to the condition number of the matrix A and explain the difference.

SOLUCIÓN — PROBLEMA 1

Problema 1 — Newton-Raphson para el recíproco de N

(a) Sea $f(x) = \frac{1}{x} - N$, de modo que $f(x) = 0 \iff x = 1/N$. Como $f'(x) = -1/x^2$, la iteración de Newton es

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\frac{1}{x_n} - N}{-\frac{1}{x_n^2}} = x_n + x_n^2 \left(\frac{1}{x_n} - N \right) = x_n + x_n - Nx_n^2 = 2x_n - Nx_n^2 = x_n(2 - Nx_n).$$

(b) Sea $\alpha = 1/N$ la raíz y $e_n = x_n - \alpha$. Sustituyendo directamente:

$$x_{n+1} - \alpha = 2x_n - Nx_n^2 - \frac{1}{N} = -N \left(x_n^2 - \frac{2x_n}{N} + \frac{1}{N^2} \right) = -N \left(x_n - \frac{1}{N} \right)^2 = -N e_n^2.$$

Es decir, **exactamente** (no solo asintóticamente)

$$e_{n+1} = -N e_n^2 \implies |e_{n+1}| = N |e_n|^2.$$

Si $x_n \rightarrow \alpha$ (es decir $e_n \rightarrow 0$) y $N \neq 0$, esto muestra que $\frac{|e_{n+1}|}{|e_n|^2} \rightarrow N$, una constante finita y no nula: por definición, la convergencia es **cuadrática** (orden 2), con constante asintótica de error $C = N$.

Verificación alternativa (vía Taylor de la función de iteración): con $g(x) = x(2 - Nx) = 2x - Nx^2$, $g'(x) = 2 - 2Nx$, $g'(\alpha) = 2 - 2N/N = 0$, y $g''(x) = -2N \neq 0$. Como $g'(\alpha) = 0$, el punto fijo es al menos de orden 2, y $e_{n+1} \approx \frac{1}{2}g''(\alpha)e_n^2 = -N e_n^2$, coincidiendo exactamente con el cálculo directo. ■

ENUNCIADO — PROBLEMA 2

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2. (a) Given n distinct data values (t_i, y_i) , $i = 1, \dots, n$, $t_i \in \mathbb{R}^1$, $y_i \in \mathbb{R}^1$, $t_1 < t_2 < \dots < t_n$, show that there exists a unique polynomial of degree at most $n - 1$ that interpolates this data. Describe a method to find this polynomial.
 - (b) Let $P_1(f; t)$ be a polynomial of degree 1 interpolating the function $f(t)$ at t_1, t_2 . If $f \in C^2[t_1, t_2]$, show that the error in linear interpolation satisfies,

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3. Consider solving the linear system of equations $Ax = b$ for an invertible matrix A and a fixed right hand side $b \neq 0$.
 - (a) Given an approximate solution $\hat{x} \neq 0$, set $\hat{b} = A\hat{x}$ and define the relative forward error (RFE) and relative backward error (RBE) for this problem.
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 - (c) Compare the condition number of the above problem to the condition number of the matrix A and explain the difference.

Problema 2 — Interpolación polinómica

(a) Existencia y unicidad. Buscamos $P(t) = \sum_{k=0}^{n-1} c_k t^k$ tal que $P(t_i) = y_i$, $i = 1, \dots, n$. Esto es un sistema lineal $n \times n$ para los coeficientes c_k , cuya matriz es la matriz de Vandermonde

$$V = (t_i^k)_{i=1, \dots, n; k=0, \dots, n-1}, \quad \det V = \prod_{1 \leq i < j \leq n} (t_j - t_i) \neq 0,$$

porque los t_i son distintos. Como $\det V \neq 0$, el sistema $Vc = y$ tiene solución única: existe un único polinomio de grado $\leq n - 1$ que interpola los datos.

Unicidad, argumento directo: si P, Q son dos polinomios de grado $\leq n - 1$ con $P(t_i) = Q(t_i) = y_i$ para $i = 1, \dots, n$, entonces $D = P - Q$ tiene grado $\leq n - 1$ y se anula en los n puntos distintos $t_1, \dots, t_n$. Un polinomio no nulo de grado $\leq n - 1$ tiene a lo más $n - 1$ raíces, así que $D \equiv 0$, es decir $P \equiv Q$.

Método constructivo (forma de Lagrange). Se define

$$P(t) = \sum_{i=1}^n y_i L_i(t), \quad L_i(t) = \prod_{\substack{j=1 \\ j \neq i}}^n \frac{t - t_j}{t_i - t_j}.$$

Cada L_i tiene grado $n - 1$, satisface $L_i(t_k) = \delta_{ik}$ (es 1 en t_i y 0 en los demás nodos), así que $P(t_k) = \sum_i y_i \delta_{ik} = y_k$ para todo k . (Alternativamente, se puede construir vía diferencias divididas de Newton, que da el mismo polinomio en forma computacionalmente más eficiente para agregar nodos.)

(b) Fórmula del error para interpolación lineal. Sea $x \in (t_1, t_2)$ fijo. Definimos la función auxiliar (de la variable t)

$$\varphi(t) = f(t) - P_1(f; t) - [f(x) - P_1(f; x)] \frac{(t - t_1)(t - t_2)}{(x - t_1)(x - t_2)}.$$

Como $f \in C^2[t_1, t_2]$ y P_1 es polinomio de grado 1, $\varphi \in C^2[t_1, t_2]$. Además:

- $\varphi(t_1) = f(t_1) - P_1(f; t_1) - 0 = 0$ (porque P_1 interpola en t_1),
- $\varphi(t_2) = 0$ análogamente,
- $\varphi(x) = [f(x) - P_1(f; x)] - [f(x) - P_1(f; x)] \cdot 1 = 0$.

Así φ se anula en los tres puntos $t_1 < x < t_2$. Por el teorema de Rolle aplicado en $[t_1, x]$ y en $[x, t_2]$, existen $\eta_1 \in (t_1, x)$, $\eta_2 \in (x, t_2)$ con $\varphi'(\eta_1) = \varphi'(\eta_2) = 0$. Aplicando Rolle de nuevo a φ' en $[\eta_1, \eta_2]$, existe $\xi = \xi(x) \in (\eta_1, \eta_2) \subset (t_1, t_2)$ con $\varphi''(\xi) = 0$.

Derivando dos veces en t (con x fijo como parámetro), y usando que $P_1''(t) \equiv 0$ por ser de grado ≤ 1 :

$$\varphi''(t) = f''(t) - [f(x) - P_1(f; x)] \cdot \frac{2}{(x - t_1)(x - t_2)}.$$

Evaluando en $t = \xi$ y usando $\varphi''(\xi) = 0$:

$$f''(\xi) = [f(x) - P_1(f; x)] \cdot \frac{2}{(x - t_1)(x - t_2)} \implies f(x) - P_1(f; x) = \frac{(x - t_1)(x - t_2)}{2} f''(\xi).$$

Renombrando la variable x como t obtenemos exactamente

$$f(t) - P_1(f; t) = (t - t_1)(t - t_2) \frac{f''(\xi(t))}{2}, \quad t_1 < t < t_2. \blacksquare$$

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 - (c) Compare the condition number of the above problem to the condition number of the matrix A and explain the difference.

4. (a) Derive the Gaussian quadrature rule using two nodes for approximating the integral $\int_{-1}^1 f(x)x^n dx$ where $n \in \mathbb{N}$ is odd.
- (b) Apply your quadrature rule to $f(x) = x^4$ and compare to the true value of the integral. Show that the error is $\mathcal{O}(n^{-3})$.
5. Assume that f satisfies $|f(x) - f(y)| < L|x - y|^\alpha$ for some $L, \alpha > 0$ and for all $x, y \in \mathbb{R}$.
- (a) Show that when $\alpha \neq 1$ we have $|f^n(x) - f^n(y)| < c \left(\frac{|x-y|}{c}\right)^{\alpha^n}$ where $c = L^{\frac{1}{1-\alpha}}$.
- (b) Assume that $f(\hat{x}) = \hat{x}$. In terms of L and α describe the set of x such that fixed point iteration is guaranteed to converge to $\hat{x}$. Consider $\alpha < 1$, $\alpha = 1$, and $\alpha > 1$.
6. Consider solving the matrix system $Ux = y$ where $U = (u_{ij})$ is a $n \times n$ upper triangular matrix by the following back-substitution algorithm:

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for  $i = n : -1 : 1$ 
    for  $j = i + 1 : n$ 
         $y_i = y_i - u_{ij} * x_j$ 
    end
     $x_i = \frac{y_i}{u_{ii}}$ 
end

```

Calculate the number of multiplications and divisions required to perform the above algorithm.

7. Consider the Initial Value Problem, $y'(t) = f(t, y(t))$, $a \leq t \leq b$, $y(a) = y_0$. Let f satisfy a Lipschitz condition in y given by $|f(t, x) - f(t, y)| \leq L|x - y|$, where L is a constant. For $i = 0, 1, \dots, n$, let the solution $y(t_i)$ and its approximation y_i generated by *Euler's method* satisfy respectively $y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(\xi_i)$ and $y_{i+1} = y_i + hf(t_i, y_i)$, where ξ_i is some point in the interval $[t_i, t_{i+1}]$, $a = t_0, t_1, \dots, t_n = b$ is a discrete set of equally spaced points with $t_i = a + ih$ and stepsize $h = \frac{b-a}{n}$. Moreover, if $y''(t)$ exists and is continuous for $a \leq t \leq b$ and that $|y''(t)| \leq M$, then prove that for each $i = 0, 1, \dots, n$:

$$|y_i - y(t_i)| \leq \frac{Mh}{2L} [e^{L(b-a)} - 1].$$

SOLUCIÓN — PROBLEMA 3

Problema 3 — Condición del problema $Ax = b$

(a) Sea $x = A^{-1}b$ la solución exacta y $\hat{x} \neq 0$ una aproximación. Definimos $\hat{b} = A\hat{x}$ (es decir, $\hat{x}$ es la solución **exacta** de un problema perturbado con lado derecho $\hat{b}$: interpretación de error hacia atrás). Se definen:

$$\text{RFE} = \frac{\|\hat{x} - x\|}{\|x\|}, \quad \text{RBE} = \frac{\|\hat{b} - b\|}{\|b\|}.$$

(b) Como $x = A^{-1}b$ y $\hat{x} = A^{-1}\hat{b}$,

$$\hat{x} - x = A^{-1}(\hat{b} - b).$$

Por lo tanto

$$\frac{\text{RFE}}{\text{RBE}} = \frac{\|A^{-1}(\hat{b} - b)\|/\|x\|}{\|\hat{b} - b\|/\|b\|} = \frac{\|A^{-1}(\hat{b} - b)\|}{\|\hat{b} - b\|} \cdot \frac{\|b\|}{\|x\|}.$$

Tomando el supremo sobre todo $\hat{b} \neq b$ (equivalentemente, sobre todo vector $d = \hat{b} - b \neq 0$), y usando la definición de norma de operador dada,

$$\sup_{\hat{b} \neq b} \frac{\|A^{-1}(\hat{b} - b)\|}{\|\hat{b} - b\|} = \sup_{d \neq 0} \frac{\|A^{-1}d\|}{\|d\|} = \|A^{-1}\|.$$

Como $\|b\|/\|x\|$ no depende de $\hat{b}$, se obtiene

$$\kappa(\text{problema}; b) = \sup_{\hat{b} \neq b} \frac{\text{RFE}}{\text{RBE}} = \|A^{-1}\| \frac{\|b\|}{\|x\|} = \|A^{-1}\| \frac{\|Ax\|}{\|x\|}.$$

(c) El número de condición de la **matriz** es $\kappa(A) = \|A\| \|A^{-1}\|$. Como $\|Ax\|/\|x\| \leq \sup_{v \neq 0} \|Av\|/\|v\| = \|A\|$ para cualquier x , se sigue

$$\kappa(\text{problema}; b) = \|A^{-1}\| \frac{\|Ax\|}{\|x\|} \leq \|A^{-1}\| \|A\| = \kappa(A).$$

La igualdad se alcanza cuando x (equivalentemente $b = Ax$) está alineado con la dirección que maximiza $\|Av\|/\|v\|$ (el vector singular derecho principal de A). **Diferencia:** $\kappa(A)$ es una cota *uniforme*, la peor amplificación posible del error relativo sobre **todos** los posibles lados derechos b ; el número de condición del problema para un b (o x) *específico* puede ser mucho menor que $\kappa(A)$ si ese x no está alineado con la dirección de máxima amplificación de A .

ENUNCIADO — PROBLEMA 4

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- (b) Apply your quadrature rule to $f(x) = x^4$ and compare to the true value of the integral. Show that the error is $\mathcal{O}(n^{-3})$.
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$$|y_i - y(t_i)| \leq \frac{Mh}{2L} [e^{L(b-a)} - 1].$$

Problema 4 — Cuadratura gaussiana de 2 puntos para peso x^n , n impar

(a) **Derivación.** Buscamos x_1, x_2, w_1, w_2 tales que

$$\int_{-1}^1 f(x)x^n dx \approx w_1 f(x_1) + w_2 f(x_2)$$

sea exacta para $f = 1, x, x^2, x^3$ (2 nodos $\Rightarrow$ 4 grados de libertad $\Rightarrow$ exactitud garantizada hasta grado $2 \cdot 2 - 1 = 3$, como en la teoría general de cuadratura gaussiana). Los momentos son $\mu_k = \int_{-1}^1 x^{n+k} dx$; como n es impar, $n+k$ es impar si k es par, y par si k es impar. Así:

$$\mu_0 = 0, \quad \mu_1 = \frac{2}{n+2}, \quad \mu_2 = 0, \quad \mu_3 = \frac{2}{n+4}.$$

Las cuatro ecuaciones de exactitud son:

$$w_1 + w_2 = 0, \quad w_1 x_1 + w_2 x_2 = \frac{2}{n+2}, \quad w_1 x_1^2 + w_2 x_2^2 = 0, \quad w_1 x_1^3 + w_2 x_2^3 = \frac{2}{n+4}.$$

De la primera, $w_2 = -w_1$. Sustituyendo en la tercera: $w_1(x_1^2 - x_2^2) = w_1(x_1 - x_2)(x_1 + x_2) = 0$; como $w_1 \neq 0$ y $x_1 \neq x_2$, se obtiene $x_2 = -x_1$ (nodos simétricos). Entonces la segunda ecuación da

$$w_1(x_1 - x_2) = 2w_1 x_1 = \frac{2}{n+2} \Rightarrow w_1 x_1 = \frac{1}{n+2},$$

y la cuarta, con $x_2^3 = -x_1^3$,

$$w_1(x_1^3 - x_2^3) = 2w_1 x_1^3 = \frac{2}{n+4} \Rightarrow w_1 x_1^3 = \frac{1}{n+4}.$$

Dividiendo: $x_1^2 = \frac{n+2}{n+4}$, es decir

$$x_1 = \sqrt{\frac{n+2}{n+4}}, \quad x_2 = -x_1, \quad w_1 = \frac{1}{(n+2)x_1} = \frac{\sqrt{n+4}}{(n+2)^{3/2}}, \quad w_2 = -w_1.$$

La regla es

$$\int_{-1}^1 f(x)x^n dx \approx w_1 [f(x_1) - f(-x_1)].$$

(b) **Aplicación a $f(x) = x^4$.** Valor exacto: $\int_{-1}^1 x^4 \cdot x^n dx = \int_{-1}^1 x^{n+4} dx = 0$, porque $n+4$ es impar (integral de una función impar en un intervalo simétrico). Valor de la regla: como f es par, $f(x_1) = f(-x_1) = x_1^4$, así

$$w_1 [f(x_1) - f(-x_1)] = w_1 \cdot 0 = 0.$$

Ambos coinciden exactamente: **el error es 0** para $f = x^4$, lo cual en particular es $O(n^{-3})$ (de hecho, más fuerte: es $O(n^{-k})$ para todo k).

Explicación estructural: como el peso x^n es una función **impar** de x (n impar) y los nodos $\{x_1, -x_1\}$ son simétricos con pesos antisimétricos $\{w_1, -w_1\}$, tanto el funcional exacto $I(f) = \int f(x)x^n dx$ como el de la regla $Q(f) = w_1[f(x_1) - f(-x_1)]$ dependen **solo** de la parte impar de f , $f_{\text{odd}}(x) = \frac{1}{2}[f(x) - f(-x)]$; toda potencia par de x (incluido x^4) contribuye automáticamente 0 a ambos lados. La regla, diseñada para ser exacta en el subespacio de polinomios impares $\{x, x^3\}$ (2 condiciones, 2 incógnitas libres x_1, w_1), no puede en cambio reproducir x^5 : para $f(x) = x^5$ el valor exacto es $\int x^{n+5} dx = \frac{2}{n+6}$ y el de la regla es $2w_1x_1^5 = \frac{2(n+2)}{(n+4)^2}$ (usando $w_1x_1 = 1/(n+2)$ y $x_1^4 = (n+2)^2/(n+4)^2$); su diferencia es

$$E(x^5) = \frac{2}{n+6} - \frac{2(n+2)}{(n+4)^2} = \frac{2[(n+4)^2 - (n+2)(n+6)]}{(n+6)(n+4)^2} = \frac{8}{(n+6)(n+4)^2} = O(n^{-3}) \quad (n \rightarrow \infty),$$

usando $(n+4)^2 - (n+2)(n+6) = (n^2 + 8n + 16) - (n^2 + 8n + 12) = 4$. Esto confirma, mediante el primer monomio impar no reproducido por la regla, exactamente el comportamiento asintótico $O(n^{-3})$ enunciado. ■

ENUNCIADO — PROBLEMA 5

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Calculate the number of multiplications and divisions required to perform the above algorithm.

7. Consider the Initial Value Problem, $y'(t) = f(t, y(t))$, $a \leq t \leq b$, $y(a) = y_0$. Let f satisfy a Lipschitz condition in y given by $|f(t, x) - f(t, y)| \leq L|x - y|$, where L is a constant. For $i = 0, 1, \dots, n$, let the solution $y(t_i)$ and its approximation y_i generated by *Euler's method* satisfy respectively $y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(\xi_i)$ and $y_{i+1} = y_i + hf(t_i, y_i)$, where ξ_i is some point in the interval $[t_i, t_{i+1}]$, $a = t_0, t_1, \dots, t_n = b$ is a discrete set of equally spaced points with $t_i = a + ih$ and stepsize $h = \frac{b-a}{n}$. Moreover, if $y''(t)$ exists and is continuous for $a \leq t \leq b$ and that $|y''(t)| \leq M$, then prove that for each $i = 0, 1, \dots, n$:

$$|y_i - y(t_i)| \leq \frac{Mh}{2L} [e^{L(b-a)} - 1].$$

Problema 5 — Condición de Hölder e iteración de punto fijo

(a) Sea $e_0 = |x - y|$ y, para $\alpha \neq 1$, $c = L^{1/(1-\alpha)}$ (bien definido y positivo porque $L > 0$). Probamos por inducción sobre $n \geq 1$ que

$$|f^n(x) - f^n(y)| < c \left(\frac{e_0}{c} \right)^{\alpha^n} =: E_n.$$

Base $n = 1$: la hipótesis del enunciado da directamente $|f(x) - f(y)| < L|x - y|^\alpha = Le_0^\alpha$. Como $c^{1-\alpha} = L$,

$$E_1 = c \left(\frac{e_0}{c} \right)^\alpha = c^{1-\alpha} e_0^\alpha = Le_0^\alpha,$$

así que la desigualdad base es exactamente la hipótesis dada.

Paso inductivo: supongamos $|f^n(x) - f^n(y)| < E_n$. Aplicando la condición de Hölder a los puntos $f^n(x)$, $f^n(y)$, y usando que $t \mapsto t^\alpha$ es estrictamente creciente para $\alpha > 0$:

$$|f^{n+1}(x) - f^{n+1}(y)| = |f(f^n(x)) - f(f^n(y))| < L |f^n(x) - f^n(y)|^\alpha < L E_n^\alpha.$$

Calculamos

$$L E_n^\alpha = L \left[c \left(\frac{e_0}{c} \right)^{\alpha^n} \right]^\alpha = L c^\alpha \left(\frac{e_0}{c} \right)^{\alpha^{n+1}}.$$

Como $L c^\alpha = c^{1-\alpha} c^\alpha = c$, esto es exactamente $c \left(\frac{e_0}{c} \right)^{\alpha^{n+1}} = E_{n+1}$. Luego $|f^{n+1}(x) - f^{n+1}(y)| < E_{n+1}$, completando la inducción. ■

(b) Sea $\hat{x}$ punto fijo, $f(\hat{x}) = \hat{x}$, y $e_n = |f^n(x) - \hat{x}| = |f^n(x) - f^n(\hat{x})|$ (ya que $f^n(\hat{x}) = \hat{x}$). Por (a), $e_n < c(e_0/c)^{\alpha^n}$ con $e_0 = |x - \hat{x}|$.

- **Caso $\alpha > 1$.** Aquí $\alpha^n \rightarrow \infty$. Si $e_0 < c$ (es decir

$e_0/c < 1$), entonces $(e_0/c)^{\alpha^n} \rightarrow 0$ (base < 1 elevada a un exponente $\rightarrow \infty$), así $e_n \rightarrow 0$: **la**

iteración converge a $\hat{x}$ siempre que el punto inicial satisfaga

$$|x - \hat{x}| < c = L^{1/(1-\alpha)}.$$

Más aún, la relación exacta de un paso $e_{n+1} < L e_n^\alpha$ (tomada de la hipótesis) muestra que la convergencia es de **orden α** (superlineal), análoga en velocidad a Newton cuando $\alpha = 2$. Si $e_0 \geq c$, el argumento no da ninguna garantía.

- **Caso $\alpha = 1$.** Entonces c no está definido ($1 - \alpha = 0$); la

hipótesis se reduce a la condición de Lipschitz usual, $|f(x) - f(y)| < L|x - y|$. Éste es el caso clásico del teorema de la aplicación contractiva: **la iteración converge a $\hat{x}$ para todo punto inicial $x \in \mathbb{R}$ si y solo si $L < 1$** (contracción global); si $L \geq 1$ no hay garantía.

- **Caso** $0 < \alpha < 1$. Ahora $\alpha^n \rightarrow 0$, así que para **cualquier**

$e_0 > 0$ fijo, $(e_0/c)^{\alpha^n} \rightarrow (e_0/c)^0 = 1$, y la cota se convierte en $e_n \lesssim c$: la cota **no** decae a 0, sino que tiende a la constante positiva c . Esto refleja que $\hat{x}$ es, en este régimen, un punto fijo típicamente **repulsor** de la dinámica mayorante (en la recursión $t \mapsto Lt^\alpha$, $t = 0$ es repulsor y $t = c$ es atractor, ya que cerca de 0, $Lt^\alpha \gg t$ para $\alpha < 1$). Por lo tanto, la condición de Hölder con $\alpha < 1$ **no garantiza convergencia** de la iteración de punto fijo a $\hat{x}$ para ningún punto inicial $x \neq \hat{x}$ — de hecho es consistente con divergencia local (ejemplo: $f(x) = \hat{x} + 2|x - \hat{x}|^{1/2}$, $\mathrm{sgn}(x - \hat{x})$ cerca de $\hat{x}$ aleja los puntos vecinos).

ENUNCIADO — PROBLEMA 6

4. (a) Derive the Gaussian quadrature rule using two nodes for approximating the integral $\int_{-1}^1 f(x)x^n dx$ where $n \in \mathbb{N}$ is odd.
- (b) Apply your quadrature rule to $f(x) = x^4$ and compare to the true value of the integral. Show that the error is $\mathcal{O}(n^{-3})$.
5. Assume that f satisfies $|f(x) - f(y)| < L|x - y|^\alpha$ for some $L, \alpha > 0$ and for all $x, y \in \mathbb{R}$.
- (a) Show that when $\alpha \neq 1$ we have $|f^n(x) - f^n(y)| < c \left(\frac{|x-y|}{c}\right)^{\alpha^n}$ where $c = L^{\frac{1}{1-\alpha}}$.
- (b) Assume that $f(\hat{x}) = \hat{x}$. In terms of L and α describe the set of x such that fixed point iteration is guaranteed to converge to $\hat{x}$. Consider $\alpha < 1, \alpha = 1$, and $\alpha > 1$.
6. Consider solving the matrix system $Ux = y$ where $U = (u_{ij})$ is a $n \times n$ upper triangular matrix by the following back-substitution algorithm:

```

for  $i = n : -1 : 1$ 
    for  $j = i + 1 : n$ 
         $y_i = y_i - u_{ij} * x_j$ 
    end
     $x_i = \frac{y_i}{u_{ii}}$ 
end

```

Calculate the number of multiplications and divisions required to perform the above algorithm.

7. Consider the Initial Value Problem, $y'(t) = f(t, y(t))$, $a \leq t \leq b$, $y(a) = y_0$. Let f satisfy a Lipschitz condition in y given by $|f(t, x) - f(t, y)| \leq L|x - y|$, where L is a constant. For $i = 0, 1, \dots, n$, let the solution $y(t_i)$ and its approximation y_i generated by *Euler's method* satisfy respectively $y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(\xi_i)$ and $y_{i+1} = y_i + hf(t_i, y_i)$, where ξ_i is some point in the interval $[t_i, t_{i+1}]$, $a = t_0, t_1, \dots, t_n = b$ is a discrete set of equally spaced points with $t_i = a + ih$ and stepsize $h = \frac{b-a}{n}$. Moreover, if $y''(t)$ exists and is continuous for $a \leq t \leq b$ and that $|y''(t)| \leq M$, then prove that for each $i = 0, 1, \dots, n$:

$$|y_i - y(t_i)| \leq \frac{Mh}{2L} [e^{L(b-a)} - 1].$$

Problema 6 — Conteo de operaciones en sustitución hacia atrás

Para cada valor fijo de i (recorriendo $i = n, n - 1, \dots, 1$), el bucle interno en j recorre $j = i + 1, \dots, n$, es decir $n - i$ valores, y cada uno ejecuta **una multiplicación** ($u_{ij} \cdot x_j$). Luego, fuera del bucle interno, se realiza **una división** (y_i / u_{ii}).

Multiplicaciones:

$$\sum_{i=1}^n (n - i) = \sum_{k=0}^{n-1} k = \frac{n(n - 1)}{2}.$$

Divisiones: una por cada $i = 1, \dots, n$, es decir n divisiones.

Total combinado (multiplicaciones + divisiones):

$$\frac{n(n - 1)}{2} + n = \frac{n(n - 1) + 2n}{2} = \frac{n(n + 1)}{2}.$$

Verificación: para $n = 1$ (sistema 1×1), la fórmula da 1 división y 0 multiplicaciones, total $1 = \frac{1 \cdot 2}{2}$. ✓. Para $n = 2$: 1 multiplicación ($i = 1, j = 2$) y 2 divisiones, total $3 = \frac{2 \cdot 3}{2}$. ✓

ENUNCIADO — PROBLEMA 7

4. (a) Derive the Gaussian quadrature rule using two nodes for approximating the integral $\int_{-1}^1 f(x)x^n dx$ where $n \in \mathbb{N}$ is odd.
- (b) Apply your quadrature rule to $f(x) = x^4$ and compare to the true value of the integral. Show that the error is $\mathcal{O}(n^{-3})$.
5. Assume that f satisfies $|f(x) - f(y)| < L|x - y|^\alpha$ for some $L, \alpha > 0$ and for all $x, y \in \mathbb{R}$.
- (a) Show that when $\alpha \neq 1$ we have $|f^n(x) - f^n(y)| < c \left(\frac{|x-y|}{c}\right)^{\alpha^n}$ where $c = L^{\frac{1}{1-\alpha}}$.
- (b) Assume that $f(\hat{x}) = \hat{x}$. In terms of L and α describe the set of x such that fixed point iteration is guaranteed to converge to $\hat{x}$. Consider $\alpha < 1, \alpha = 1$, and $\alpha > 1$.
6. Consider solving the matrix system $Ux = y$ where $U = (u_{ij})$ is a $n \times n$ upper triangular matrix by the following back-substitution algorithm:

```

for  $i = n : -1 : 1$ 
    for  $j = i + 1 : n$ 
         $y_i = y_i - u_{ij} * x_j$ 
    end
     $x_i = \frac{y_i}{u_{ii}}$ 
end

```

Calculate the number of multiplications and divisions required to perform the above algorithm.

7. Consider the Initial Value Problem, $y'(t) = f(t, y(t))$, $a \leq t \leq b$, $y(a) = y_0$. Let f satisfy a Lipschitz condition in y given by $|f(t, x) - f(t, y)| \leq L|x - y|$, where L is a constant. For $i = 0, 1, \dots, n$, let the solution $y(t_i)$ and its approximation y_i generated by *Euler's method* satisfy respectively $y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(\xi_i)$ and $y_{i+1} = y_i + hf(t_i, y_i)$, where ξ_i is some point in the interval $[t_i, t_{i+1}]$, $a = t_0, t_1, \dots, t_n = b$ is a discrete set of equally spaced points with $t_i = a + ih$ and stepsize $h = \frac{b-a}{n}$. Moreover, if $y''(t)$ exists and is continuous for $a \leq t \leq b$ and that $|y''(t)| \leq M$, then prove that for each $i = 0, 1, \dots, n$:

$$|y_i - y(t_i)| \leq \frac{Mh}{2L} [e^{L(b-a)} - 1].$$

Problema 7 — Cota de error global del método de Euler

Sea $e_i = |y_i - y(t_i)|$. Restando la expansión de Taylor exacta y la fórmula de Euler:

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2} y''(\xi_i), \quad y_{i+1} = y_i + hf(t_i, y_i),$$

$$y_{i+1} - y(t_{i+1}) = [y_i - y(t_i)] + h[f(t_i, y_i) - f(t_i, y(t_i))] - \frac{h^2}{2} y''(\xi_i).$$

Tomando valor absoluto, la desigualdad triangular, la condición de Lipschitz $|f(t_i, y_i) - f(t_i, y(t_i))| \leq L|y_i - y(t_i)|$, y $|y''(\xi_i)| \leq M$:

$$e_{i+1} \leq (1 + hL) e_i + \frac{h^2 M}{2}, \quad e_0 = 0 \quad (\text{condición inicial exacta}).$$

Resolviendo la recursión. Ésta es una recursión lineal de primer orden. Iterando (o por inducción, verificando que la fórmula satisface la recursión con igualdad como cota superior):

$$e_i \leq \frac{h^2 M}{2} \sum_{k=0}^{i-1} (1 + hL)^k = \frac{h^2 M}{2} \cdot \frac{(1 + hL)^i - 1}{(1 + hL) - 1} = \frac{hM}{2L} [(1 + hL)^i - 1].$$

(Verificación por inducción de esta suma geométrica: para $i = 0$ ambos lados son 0; si $e_i \leq \frac{hM}{2L} [(1 + hL)^i - 1]$, entonces $e_{i+1} \leq (1 + hL)e_i + \frac{h^2 M}{2} \leq \frac{hM}{2L} [(1 + hL)^{i+1} - 1]$, cerrando la inducción.)

Cota final. Usando la desigualdad elemental $1 + z \leq e^z$ (válida para todo $z \in \mathbb{R}$, en particular $z = hL > 0$), se tiene $(1 + hL)^i \leq e^{ihL} = e^{L(t_i - a)} \leq e^{L(b-a)}$ para $t_i \leq b$. Por lo tanto

$$|y_i - y(t_i)| = e_i \leq \frac{hM}{2L} [e^{L(b-a)} - 1], \quad i = 0, 1, \dots, n.$$

Verificación de casos límite: si $M = 0$ (es decir $y'' \equiv 0$, la solución exacta es afín en t , reproducida exactamente por Euler en ese caso especial... más precisamente si y es lineal el error de truncamiento local es cero) la cota da $e_i = 0$ para todo i , consistente. Si $h \rightarrow 0$ con $n = (b-a)/h \rightarrow \infty$, la cota $\rightarrow 0$ linealmente en h , confirmando que Euler es un método de **orden 1** (convergencia global $O(h)$). ■